

Commutator

Def. Let G be a group. Given $x, y \in G$, define

$$[x, y] \stackrel{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the commutator of x and y . And, given two subset A and B ,

$$[A, B] \stackrel{\text{def}}{=} \langle [a, b] \mid a \in A, b \in B \rangle$$

Particularly, $G' \stackrel{\text{def}}{=} [G, G]$ is called the derived subgroup.

Prop. G' char G , (so $G' \trianglelefteq G$).

Proof. Given $\sigma \in \text{Aut}(G)$ and $[x, y]$ for arbitrary $x, y \in G$,

$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

$$\text{So, } \sigma[[G, G]] = [G, G].$$

Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H, G] \leq H$

Proof. $H \trianglelefteq G \Leftrightarrow$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$$[H, G] \leq H \Leftrightarrow \text{for all } g \in G \text{ and } h \in H, [h, g] = h^{-1}g^{-1}hg \in H.$$

So proved. $\square$

Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x, y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$. $\square$

Note: G' is the smallest normal subgroup the quotient is abelian.

In other say, G/G' is the largest abelian quotient group.

$$G' = \bigcap_{N \trianglelefteq G, G/N \text{ is abelian}} N$$

Prop. If $\phi: G \rightarrow A$ is a Homomorphism into an abelian group A , then $G' \leq \ker \phi$.

Proof. By the first Isomorphism Theorem, $G/\ker \phi \cong \text{Im } \phi \leq A$.

Since A is abelian, so $\text{Im } \phi$ is also abelian. Therefore, $G' \leq \ker \phi$. $\square$

Alternatively, given $x, y \in G$, $\phi([x, y]) = \phi(x)^{-1}\phi(y)^{-1}\phi(x)\phi(y) = 1_A$. $\square$

Homomorphism decomposition.

Let $\varphi: G \rightarrow A$ be a Group Homomorphism into abelian group A .
By the First-Isomorphism theorem, $G/\ker \varphi \cong \text{Im } \varphi \leq A$, so $G' \leq \ker \varphi$.
Therefore, $\psi: G/G' \rightarrow A; \alpha G' \mapsto \varphi(\alpha)$ is well-defined because

$$\alpha_1 G' = \alpha_2 G' \Leftrightarrow \alpha_1^{-1} \alpha_2 \in G' \Rightarrow \alpha_1^{-1} \alpha_2 \in \ker \varphi \Leftrightarrow \varphi(\alpha_1^{-1} \alpha_2) = e \Leftrightarrow \varphi(\alpha_1) = \varphi(\alpha_2)$$

Now, the onto Homomorphism, $\pi: G \rightarrow G/G'; g \mapsto gG'$ satisfies:

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & A \\ \pi \downarrow \wr & \searrow \varphi & \\ G/G' & \xrightarrow{\psi} & A \end{array} \quad \varphi = \psi \circ \pi.$$

And,

$$\ker \varphi = \varphi^{-1}[\{e\}] = \pi^{-1}[\psi^{-1}[\{e\}]] = \pi^{-1}[\ker \psi].$$

Commutator of Typical Subgroups.

D_{2n} , the dihedral of order $2n$.

Prop. $D'_{2n} = \langle r^2 \rangle \cong \begin{cases} C_n & \text{if } n \text{ odd} \\ C_{\frac{n}{2}} & \text{if } n \text{ even} \end{cases}$

S_n , the symmetric group on n letters

Prop. $S_n' = A_n$.

Proof. Since $S_n/A_n \cong C_2$, abelian, so $S_n' \leq A_n$.

Debris,

$$[S_3, S_3] = \langle [a, b] \mid a, b \in S_3 \rangle$$

$$[(1\ 2), (1\ 3)] = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 3\ 2)(1\ 3\ 2) = (1\ 2\ 3)$$

$$[(1\ 2), (2\ 3)] = (1\ 2)(2\ 3)(1\ 2)(2\ 3) = (1\ 2\ 3\ 1)(1\ 2\ 3\ 1) = (1\ 3\ 2)$$

$$[(1\ 3), (2\ 3)] = (1\ 3)(2\ 3)(1\ 3)(2\ 3) = (3\ 2\ 1)(3\ 2\ 1) = (1\ 2\ 3)$$

$$[(1\ 2), (1\ 2\ 3)] = (1\ 2)(1\ 2\ 3)^{-1}(1\ 2)(1\ 2\ 3) = (1\ 2)(3\ 2\ 1)(1\ 2)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 3), (1\ 2\ 3)] = (1\ 3)(1\ 2\ 3)^{-1}(1\ 3)(1\ 2\ 3) = (1\ 3)(3\ 2\ 1)(1\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(2\ 3), (1\ 2\ 3)] = (2\ 3)(1\ 2\ 3)^{-1}(2\ 3)(1\ 2\ 3) = (2\ 3)(3\ 2\ 1)(2\ 3)(1\ 2\ 3) = (1\ 3\ 2)$$

$$[(1\ 2\ 3), (1\ 3\ 2)] = \text{id}.$$

$$\text{Therefore, } [S_3, S_3] = \langle (1\ 2\ 3), (1\ 3\ 2) \rangle = \{\text{id}, (1\ 2\ 3), (1\ 3\ 2)\} = A_3.$$

Alternatively, since A_3 is normal subgroup of S_3 and $S_3/A_3 \cong C_2$, so $[S_3, S_3] \subseteq A_3$. Conversely, since $(1\ 3\ 2) \in A_3$ and $(1\ 3\ 2) = [(1\ 2), (2\ 3)] \in [S_3, S_3]$. Therefore $[S_3, S_3] \supseteq A_3 = \langle (1\ 3\ 2) \rangle$.

Since $S_n = \langle (1\ 2), (1\ 2\ 3\ \dots\ n) \rangle$, because:

$$(1\ 2\ \dots\ n)(1\ 2)(n\ \dots\ 2\ 1) = (2\ 3)$$

$$(1\ 2\ \dots\ n)(2\ 3)(n\ \dots\ 2\ 1) = (3\ 4)$$

$$\vdots$$
$$(1\ 2\ \dots\ n)(n-2\ n-1)(n\ \dots\ 2\ 1) = (n-1\ n)$$

and for any $1 \leq i \leq n$

$$(i\ i+1)(i+1\ i+2)(i\ i+1) = (i\ i+2)$$

$$(i\ i+2)(i+2\ i+3)(i\ i+2) = (i\ i+3)$$

$$\vdots$$
$$(i\ i+s)(i+s\ i+s+1)(i\ i+s) = (i\ i+s+1), \text{ inductively,}$$

$\langle (1\ 2), (1\ 2\ \dots\ n) \rangle$ contains all transpositions in S_n , so it generates S_n .

Now,

$$S_n' = [S_n, S_n] =$$

Γ_{A_n} , the alternating group.

Γ_8 . The Quaternion.